



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

Mathematics Extension 1

Year 11 Yearly Examination, 2022

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- The HSC Reference Sheet will be provided.
- For questions in Section II, show relevant mathematical reasoning and/or calculations

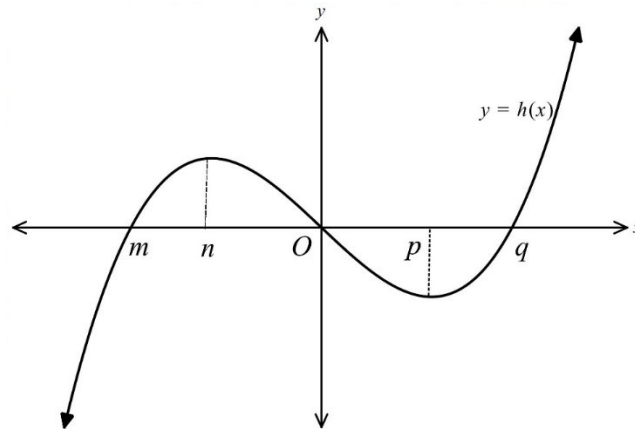
	Marks
Section I: Multiple Choice	/10
Section II: Question 11	/15
Section II: Question 12	/14
Section II: Question 13	/16
Section II: Question 14	/15
Total	/70

BLANK PAGE

Section I: Multiple Choice (10 Marks)

Answer questions 1 - 10 on the **Multiple-choice Answer sheet** provided.
Allow about 15 minutes for this section.

1. The graph of a cubic function $y = h(x)$ is shown below. The function has 1
 x -intercepts at $x = m$, $x = 0$ and $x = q$, and has turning points where $x = n$
and $x = p$.



Which of the following restrictions on the domain would allow the function $y = h(x)$ to have an inverse function $y = h^{-1}(x)$?

- (A) $[m, 0]$
 (B) $[m, q]$
 (C) $[0, \infty)$
 (D) $[n, p]$
2. What is the highest degree of the remainder of a sixth degree polynomial when 1
it is divided by a fourth degree polynomial?
- (A) zero
 (B) one
 (C) two
 (D) three

3. Which of the following is a true statement? 1

(A) $\sin 3x \sin 4y = \frac{1}{2} [\cos(3x - 4y) - \cos(3x + 4y)]$

(B) $\sin 3x \sin 4y = \frac{1}{2} [\cos(3x + 4y) - \sin(3x - 4y)]$

(C) $\sin 3x \sin 4y = \frac{1}{2} [\cos(3x + 4y) - \cos(3x - 4y)]$

(D) $\sin 3x \sin 4y = \frac{1}{2} [\cos(3x - 4y) - \sin(3x + 4y)]$

4. Which one of the following is the inverse function of $f(x) = x^2 - 4x + 2$, for $x \geq 2$? 1

(A) $f^{-1}(x) = 2 + \sqrt{x + 2}$

(B) $f^{-1}(x) = 2 - \sqrt{x + 2}$

(C) $f^{-1}(x) = 2 + \sqrt{x - 2}$

(D) $f^{-1}(x) = 2 - \sqrt{x - 2}$

5. Let α and β be the roots of the polynomial $P(x) = 3x^2 - x + 2$. What is the value of $\frac{1}{\alpha} + \frac{1}{\beta}$? 1

(A) $-\frac{2}{3}$

(B) $-\frac{1}{2}$

(C) $\frac{1}{2}$

(D) $\frac{2}{3}$

6. If $A + B = 45^\circ$ and $\tan A = 2$, what is the value of $\tan B$? 1

(A) $-\frac{1}{2}$

(B) $-\frac{1}{3}$

(C) $\frac{1}{3}$

(D) $\frac{1}{2}$

7. If $\sin \theta = \frac{1}{\sqrt{3}}$, and $90^\circ < \theta < 180^\circ$, what is the exact value of $\sin 2\theta$? 1

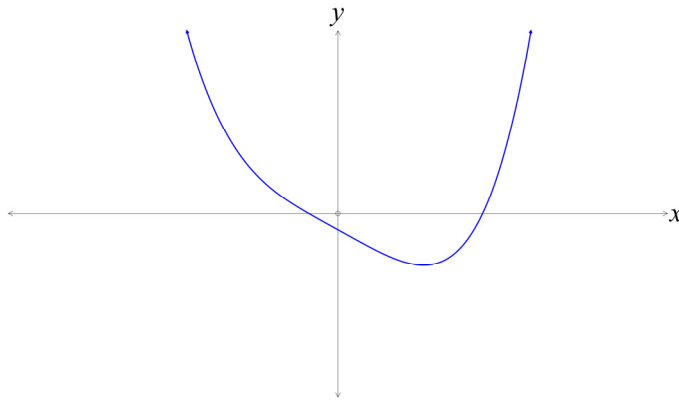
(A) $-\frac{2\sqrt{2}}{3}$

(B) $-\frac{2\sqrt{3}}{3}$

(C) $\frac{2\sqrt{2}}{3}$

(D) $\frac{2\sqrt{3}}{3}$

8. The graph of polynomial $y = P(x)$ is given below. 1



Which one of the following could be the equation of $P(x)$?

(A) $P(x) = x^3 - x - 1$

(B) $P(x) = x^4 - x - 1$

(C) $P(x) = x^6 - x + 1$

(D) $P(x) = -x^8 - x - 1$

9. $P(x) = x^3 + 3x^2 - kx + 4$ has a remainder of k when divided by $x - 2$. What is the value of k ? 1

- (A) -1
(B) 2
(C) 5
(D) 8

10. What is the domain of the function $y = 3 \sin^{-1}(x)$? 1

- (A) $-\frac{1}{3} \leq x \leq \frac{1}{3}$
(B) $0 \leq x \leq \frac{1}{3}$
(C) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
(D) $-1 \leq x \leq 1$

END OF SECTION I

Section II (60 Marks)**Start a new writing booklet for each question.**

Attempt Questions 11 - 14.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculation.

Allow about 1 hour and 45 minutes for this section.

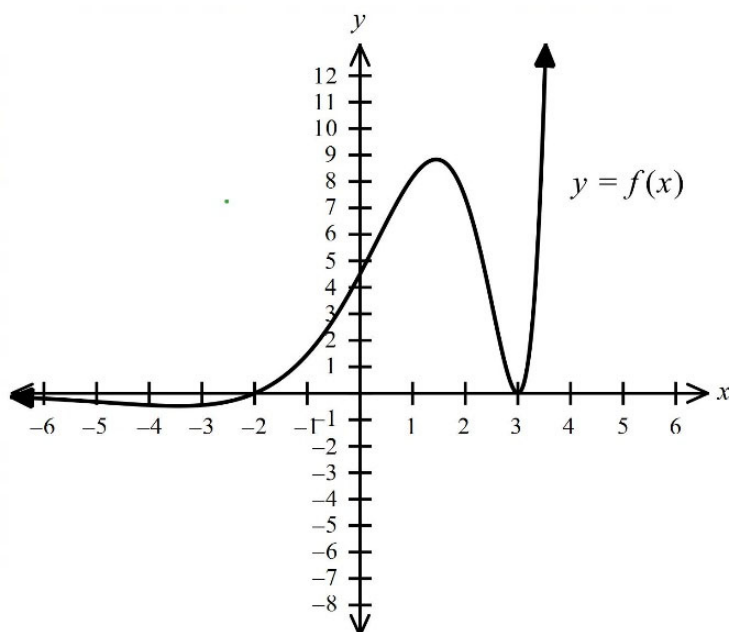
Question 11 (15 marks)**Start a new writing booklet.**

- (a) Find the exact value of $\cos 75^\circ \sin 45^\circ$. 3
- (b) Solve the inequality $|x - 4| \geq 2$. 2
- (c) Show that $\frac{1}{1 + \sqrt{2} \sin x} + \frac{1}{1 - \sqrt{2} \sin x} = 2 \sec 2x$. 3
- (d) If $t = \tan \frac{x}{2}$, show that $2 \cos x - 5 \sin x = -2 \left(\frac{t^2 + 5t - 1}{t^2 + 1} \right)$. 2
- (e) Solve $3x^2 - 8x - 7 \geq 2x^2 - 2x + 9$. 2

This question is continued on the next page.

- (f) Use Insert A to answer this question. Draw your graphs directly on the insert, then put the insert inside your Question 11 writing booklet.

The graph of $y = f(x)$ is shown below.



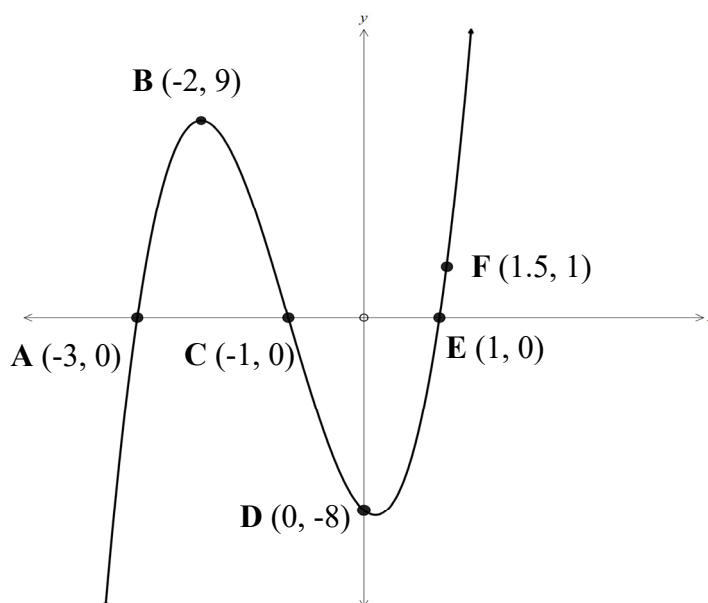
- (i) Draw a sketch of $y = |f(x)|$. Clearly indicate on your sketch all key values. Use Insert A. 1
- (ii) Draw a sketch of $y = \frac{1}{f(x)}$. Clearly indicate on your sketch all key values and possible asymptotes. Use Insert A. 2

END OF QUESTION 11

Question 12 (14 marks)**Start a new writing booklet.**

- (a) A particle is moving along the x -axis such that its displacement (x metres) from a fixed-point O after t seconds is given by $x = 2t^3 - 12t^2$.
- (i) Sketch the graph of displacement x against time t , for $t \geq 0$. Clearly indicate on your graph all co-ordinate intercepts. **2**
- (ii) Find the velocity of the particle at 6 seconds. **1**
- (b) Use **Insert B** to answer this question. Draw your graph directly on the insert, then put the insert inside your Question 12 writing booklet.

The graph below shows $y = f(x)$. **3**

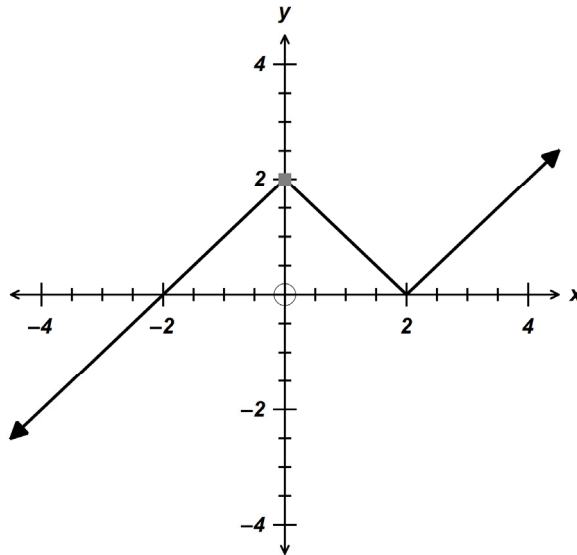


Sketch the graph of $y^2 = f(x)$ on **Insert B**. Indicate on your graph the new locations of points **A**, **B**, **C**, **D**, **E**, **F**, clearly stating their coordinates, where appropriate.

This question is continued on the next page.

- (c) Express the polynomial $P(x) = x^3 + 4x^2 - 11x - 30$ as the product of three linear factors. **3**
- (d) Use **Insert C** to answer this question. Draw your graph directly on the insert, then put the insert inside your Question 12 writing booklet.

The graph below shows $y = f(x)$ where $f(x) = \begin{cases} x+2 & \text{for } x < 0 \\ |x-2| & \text{for } x \geq 0 \end{cases}$.



- (i) Sketch the graph of $y = f(-x)$. Clearly indicate all intercepts on the co-ordinate axes. Use **Insert C**. **2**
- (ii) Sketch the graph of $y = f(x) + f(-x) + 1$, carefully showing any intercepts on the co-ordinate axes. Use **Insert C**. **3**

END OF QUESTION 12

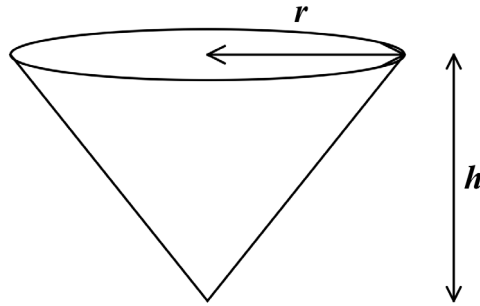
Question 13 (16 marks)**Start a new writing booklet.**

- (a) Show that $\sec(\tan^{-1} p) = \sqrt{1 + p^2}$. **2**
- (b) Solve $P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24 = 0$, given that $P(x) = 0$ has a root of multiplicity 3. **3**
- (c) The polynomial equation $2x^3 - 4x^2 + 7x - 9 = 0$ has roots α , β and γ . Find the value of $(\alpha + 1)(\beta + 1)(\gamma + 1)$. **2**
- (d) A metal is heated to a temperature of 180° and placed in a room to cool. The temperature of the room is a constant 20° . The temperature of the metal, T , changes such that $\frac{dT}{dt} = -k(T - 20)$, where k is a constant and t is the time in minutes after the metal has been placed in the room.
- (i) Show that $T = 20 + Ae^{-kt}$, where A is a constant, satisfies **1**

$$\frac{dT}{dt} = -k(T - 20).$$
- (ii) The metal is initially cooling at a rate of 6° per minute. Show that **2**
 $k = 0.0375$.

This question is continued on the next page.

- (e) A conical paper cup is equal in radius and height, where $r = h$. The cup is being filled with water so that the water level rises at a rate of 2cm/sec. Find the rate that water is being poured into the cup in cm^3/sec when the water level is 2cm. 3



- (f) A curve is defined by the parametric equations:

$$x = 3\cos(t) + 4 \text{ and } y = 3\sin(t) - 5$$

- (i) The point $P(4, -2)$ lies on the curve. Find the value of t within the domain $[0, 2\pi]$ which corresponds to the point P . 1
- (ii) Find the Cartesian equation which describes the curve. 2

END OF QUESTION 13

Question 14 (15 marks)**Start a new writing booklet.**

- (a) The piecewise function $f(x)$ is defined below. 2

$$f(x) = \begin{cases} x-1, & x \leq 0 \\ x+1, & 0 < x \leq 3 \\ x+10 & x > 3 \end{cases}$$

Find the domain of $f^{-1}(x)$.

- (b) Solve $\frac{x^2-2}{x-2} \geq 1$. 3

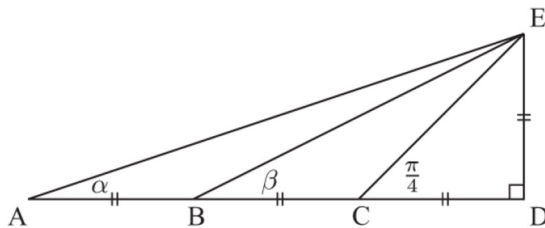
- (c) Consider the function $f(x) = \cos^{-1}(x-1) + 2\sin^{-1}(x-2)$.

- (i) Find the exact value of k given $f(1) = k\pi$. 1

- (ii) Find the domain of $f(x)$. 2

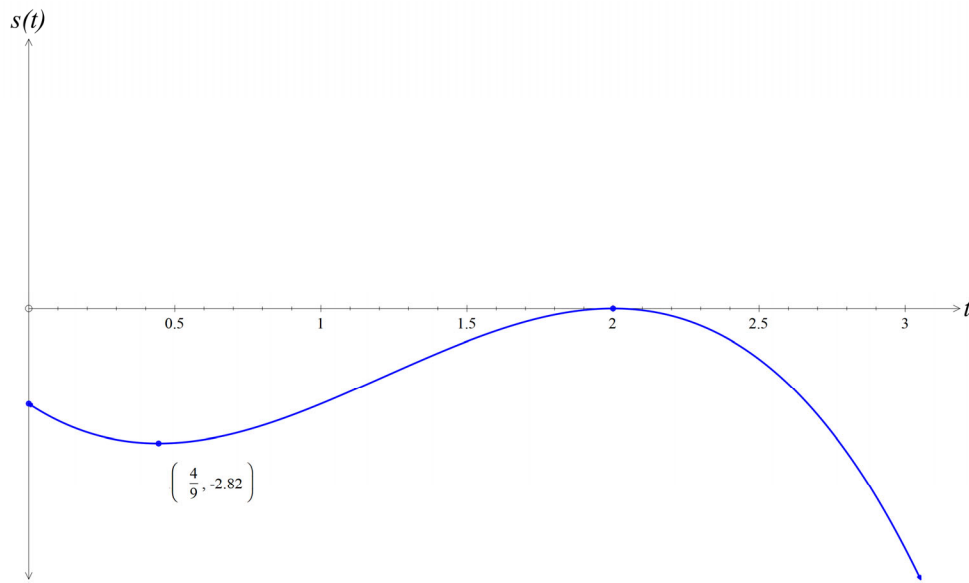
- (d) Find the exact value of $\sin\left(\arctan\frac{4}{3} + 2\arctan\frac{1}{2}\right)$. Show all working. 3

- (e) Prove that, in the diagram below, $\alpha + \beta = \frac{\pi}{4}$. 2



This question is continued on the next page.

- (f) The displacement $s(t)$ of a particle from the origin at a time t seconds is given by the graph below. Sketch the velocity function of the particle, clearly indicating all key values. 2



END OF EXAM

Y11 Extension 1 Yearly Exam 2022 Term 3 – Solutions

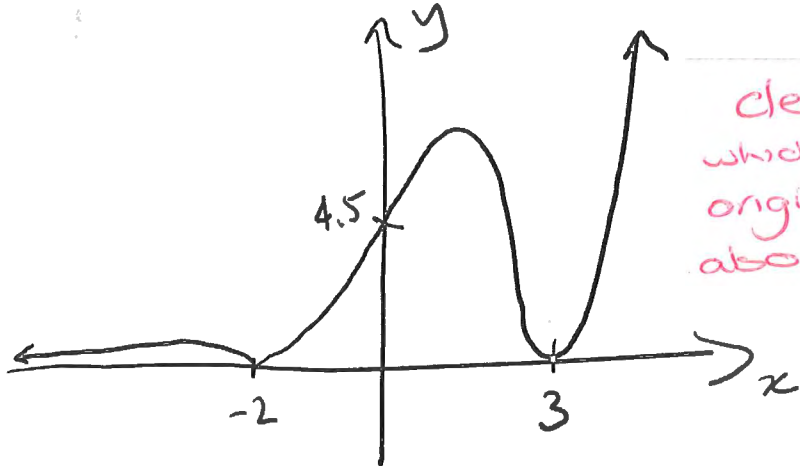
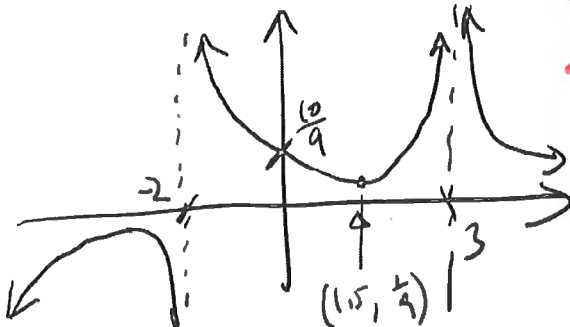
Section I – Multiple Choice

1	D As that is the only section where the function is monotonically decreasing.	
2	D Because if the remainder is of degree 3, it can no longer be divided by a divisor of degree 4.	
3	A By definition. Refer to the reference sheet.	
4	A $y = (x - 2)^2 - 4 + 2$ $y + 2 = (x - 2)^2$ $x - 2 = \sqrt{y + 2}$ $x = 2 + \sqrt{y + 2}$ $f^{-1}(x) = 2 + \sqrt{x + 2}$	
5	C $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$ $\alpha + \beta = \frac{1}{2}$ $\alpha\beta = \frac{2}{3}$ $\therefore \frac{\alpha + \beta}{\alpha\beta} = \frac{1}{2}$	
6	B $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ $\tan 45^\circ = \frac{2 + \tan B}{1 - 2 \tan B}$ $1 = \frac{2 + \tan B}{1 - 2 \tan B}$ $\tan B = -\frac{1}{3}$	

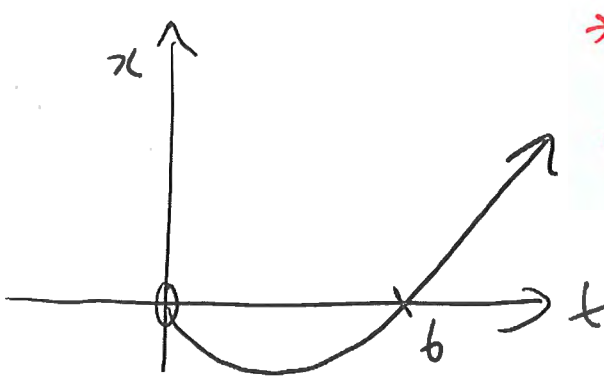
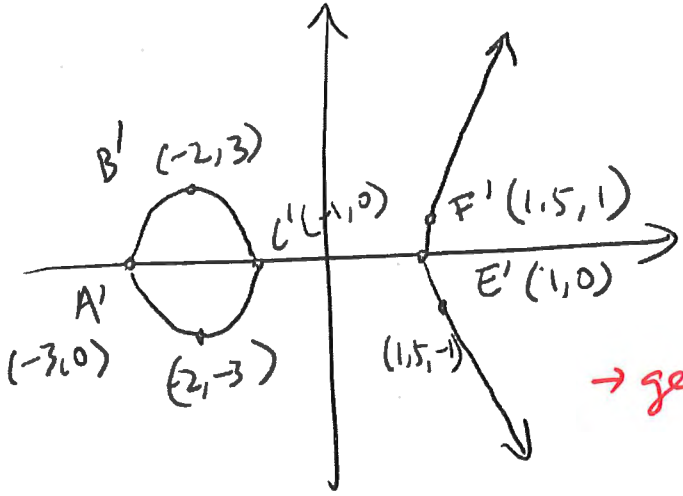
7	A $\because \sin \theta = \frac{1}{\sqrt{3}}, 90^\circ < \theta < 180^\circ,$ $\cos \theta = -\frac{\sqrt{2}}{\sqrt{3}}$ $\sin 2\theta = 2 \cos \theta \sin \theta$ $= 2 \left(\frac{1}{\sqrt{3}} \right) \left(-\frac{\sqrt{2}}{\sqrt{3}} \right)$ $= -\frac{2\sqrt{2}}{3}$	
8	B As the power of the polynomial must be even, the leading coefficient must be positive, and the y-intercept must be negative.	
9	D $P(2) = (2)^3 + 3(2)^2 - k(2) + 4 = k$ $k = 24 - 3k$ $k = 8$	
10	D As the domain of $\sin^{-1}(x)$ is $-1 \leq x \leq 1$. The 3 stretches the graph vertically by a factor three and will affect the range of the function, but not the domain.	

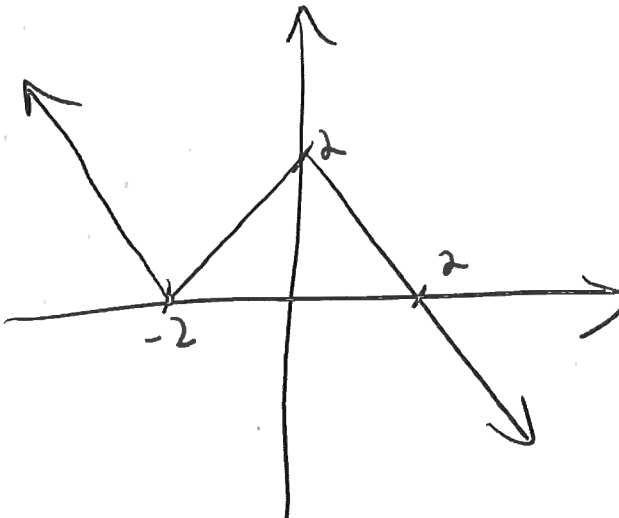
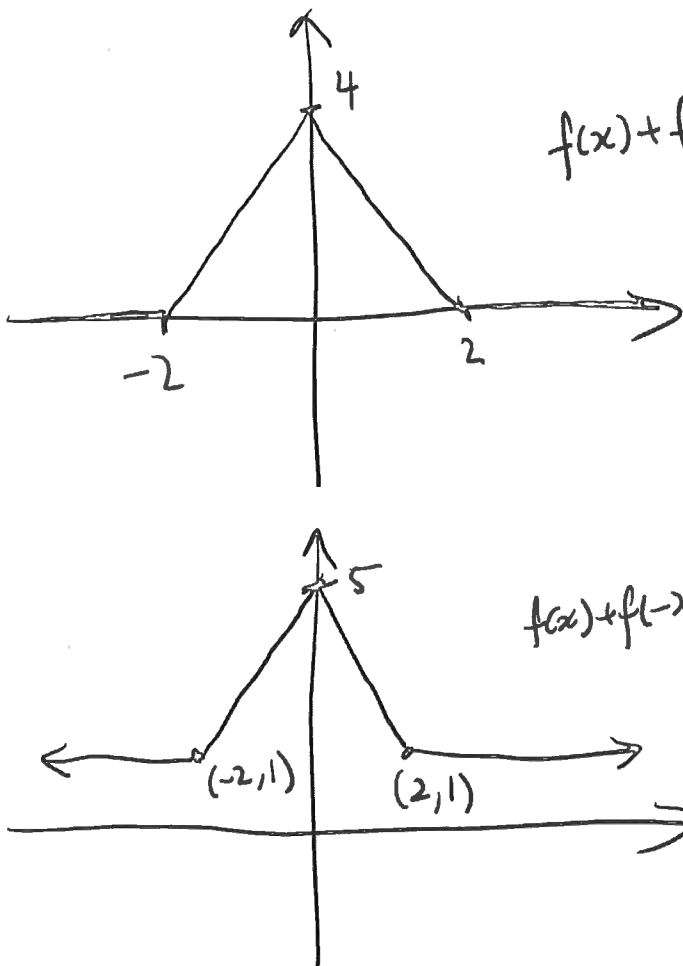
Section II- Question 11

(a)	$\cos 75^\circ \sin 45^\circ = \cos(45^\circ + 30^\circ) \times \frac{1}{\sqrt{2}}$ $= \frac{1}{\sqrt{2}} (\cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ)$ $= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2} \right)$ $= \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}-1}{2\sqrt{2}} \right)$ $= \frac{\sqrt{3}-1}{4}$ <i>Method 2.</i> $\cos 75^\circ \sin 45^\circ = \frac{1}{2} [\sin(75^\circ + 45^\circ) - \sin(75^\circ - 45^\circ)]$ <i>refer to triangles on reference sheet for exact values.</i>	3 marks
(b)	$ x-4 =2$ $x-4=2, x=6$ $x-4=-2, x=2$ $\therefore x \leq 2, x \geq 6$ <i>Method 2 Solve 2 cases</i> ① $x-4=2$ ② $-(x-4)=2$ $\rightarrow$ cannot be written as $6 \leq x \leq 2$.	2 marks
(c)	$\frac{1}{1+\sqrt{2}\sin x} + \frac{1}{1-\sqrt{2}\sin x} = \frac{(1-\sqrt{2}\sin x) + (1+\sqrt{2}\sin x)}{1-2\sin^2 x}$ $= \frac{2}{1-2(1-\cos^2 x)}$ $= \frac{2}{1-2+2\cos^2 x}$ $= \frac{2}{2\cos^2 x - 1}$ $= \frac{2}{\cos 2x}$ $= 2\sec 2x$ <i>recognise denominator is difference of 2 squares</i> <i>Consider the number of marks allocated as an indication of amount of working to write.</i>	3 marks
(d)	$2\cos x - 5\sin x = 2\left(\frac{1-t^2}{1+t^2}\right) - 5\left(\frac{2t}{1+t^2}\right)$ $= \frac{2-2t^2-10t}{1+t^2}$ $= -2\left(\frac{t^2+5t-1}{1+t^2}\right)$ <i>Use reference sheet.</i>	2 marks
(e)	$3x^2 - 8x - 7 \geq 2x^2 - 2x + 9$ $x^2 - 6x - 16 \geq 0$ $(x+2)(x-8) \geq 0$ $x \leq -2, x \geq 8$ <i>test point or draw parabola</i>	2 marks

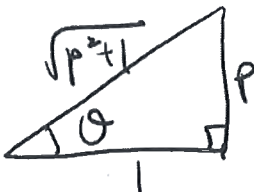
(f)	(i)  clearly indicate which parts of the original graph are also in new graph,	1 marks
	(ii)  • show new y intercepts • points of intersection of graphs • asymptotes.	2 marks

Question 12

(a)	(i) $x = 2t^2(t-6)$  → time t can not be negative. → Otherwise generally well done	2 marks
	(ii) $v = \frac{dx}{dt} = 6t^2 - 24t$ $v = 6(6)^2 - 24(6)$ $= 72 \text{ ms}^{-1}$ → Generally well Done	1 mark
(b)	 → generally well Done.	3 marks
(c)	$P(-2) = (-2)^3 + 4(-2)^2 - 11(-2) - 30 = 0$ $\therefore P(x) = (x+2)(x^2 + 2x - 15)$ $= (x+2)(x+5)(x-3)$ → Generally well Done. → Some students used long division → very time consuming!	3 marks

(d)	(i)  → Generally well Done	2 marks
	(ii)  → Poorly Done . → Use table of value for the y-values of $f(x)$ and $f(-x)$ to figure the new y-value out.	3 marks

Question 13

(a)	Let $\theta = \tan^{-1} p$  $\sec \theta = \frac{1}{\cos \theta}$ $\therefore \cos \theta = \frac{1}{\sqrt{1+p^2}}$ $\therefore \sec \theta = \sqrt{1+p^2}$	2 marks
(b)	$P'(x) = 4x^3 - 9x^2 - 12x + 28$ $P''(x) = 12x^2 - 18x - 12$ $0 = 12x^2 - 18x - 12$ $0 = 6(2x^2 - 3x - 2)$ $0 = (2x+1)(x-2)$ $x = -\frac{1}{2}, x = 2$ $P(2) = 0$ $\therefore x^4 - 3x^3 - 6x^2 + 28x - 24 = (x-2)^3(x+3)$ $x = 2, x = -3$	3 marks
(c)	$(\alpha+1)(\beta+1)(\gamma+1)$ $= (\alpha\beta + \alpha + \beta + 1)(\gamma+1)$ $= \alpha\beta\gamma + \alpha\beta + \alpha\gamma + \alpha + \beta\gamma + \beta + \gamma + 1$ $= (\alpha\beta\gamma) + (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma) + 1$ $= \left(\frac{9}{2}\right) + \left(\frac{7}{2}\right) + (2) + 1$ $= 11$	2 marks
(d)	(i) $T = 20 + Ae^{-kt}$ $\frac{dT}{dt} = -kAe^{-kt}$ $= -k(Ae^{-kt} + 20 - 20)$ $= -k(T - 20)$	1 mark

You need to include a diagram with all the information to support your answer.

Generally well done.
Long division is another method.
You need to solve, so don't stop in the factorised form

Generally well done.
Some mistakes in expanding.

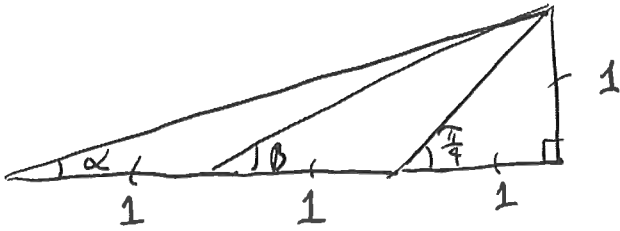
It's a show question so you need to include all the necessary steps.

	(ii) $-6 = -k(180 - 20)$ $-6 = -k \times 160$ $k = \frac{6}{160} = 0.0375$	2 marks Most missed that $\frac{dV}{dt} = -6$. Some used average rate of change, not instantaneous.
(e)	$V = \frac{\pi r^2 h}{3} \therefore r = h, V = \frac{\pi r^3}{3}$ $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ $= \pi r^2 \times 2$ $= \pi(2)^2 \times 2$ $= 25.1 \text{ cm}^3 / \text{sec}$	3 marks It was okay Some used the wrong formula show all steps.
(f)	(i) $4 = 3 \cos t + 4$ $0 = \cos t, t = \frac{\pi}{2}, t = \frac{3\pi}{2}$ $-2 = 3 \sin t - 5$ $1 = \sin t, t = \frac{\pi}{2}$ $\therefore t = \frac{\pi}{2}$	1 mark ← That produces 2 solutions in the given domain, you need to give them both $\sin t$ then limits the answer to one.
	(ii) $\cos t = \frac{x-4}{3}, \sin t = \frac{y+5}{3}$ $\sin^2 t + \cos^2 t = 1$ $\left(\frac{x-4}{3}\right)^2 + \left(\frac{y+5}{3}\right)^2 = 1$ $(x-4)^2 + (y+5)^2 = 9$	2 marks It was generally well done. Some forgot to square 3.

Question 14

(a)	Range for $f(x)$ is: For $x \leq 0, y \in (-\infty, -1]$ For $0 < x \leq 3, y \in (1, 4]$ For $x > 3, y \in (13, \infty)$ $\therefore$ for $f^{-1}(x)$, $x \in (-\infty, -1] \cup (1, 4] \cup (13, \infty)$ $x \leq -1, 1 < x \leq 4, x > 13$	2 marks
(b)	$\frac{x^2 - 2}{x - 2} = 1$ Key values: $x = 2$ $x^2 - 2 = x - 2$ $x^2 - x = 0$ $x(x - 1) = 0 \therefore x = 0, x = 1$ Testing values: $x = -2$: $\frac{(-2)^2 - 2}{-2 - 2} = -\frac{1}{2} \geq 1$ is false $x = 0.5$ $\frac{(0.5)^2 - 2}{0.5 - 2} = \frac{7}{6} \geq 1$ is true $x = 1.5$ $\frac{(1.5)^2 - 2}{1.5 - 2} = -\frac{1}{2} \geq 1$ is false $x = 5$ $\frac{(5)^2 - 2}{5 - 2} = \frac{23}{3} \geq 1$ is true $\therefore 0 \leq x \leq 1, x > 2$ METHOD 2	3 marks

	$(x^2 - 2)(x - 2) \geq (x - 2)^2$ <i>Keep in factorised form.</i> $(x^2 - 2)(x - 2) - (x - 2)^2 \geq 0$ <i>Do not expand.</i> $(x - 2)[(x^2 - 2) - (x - 2)] \geq 0$ $(x - 2)(x^2 - x) \geq 0$ <i>test values or draw cubic.</i> $x(x - 2)(x - 1) \geq 0$ <i>do not divide by x as the solution $x = 0$ is then lost.</i> $\therefore 0 \leq x \leq 1, x > 2$	
(c)	(i) $f(1) = \cos^{-1}(0) + 2\sin^{-1}(-1)$ <i>$\sin^{-1}(-x) = -\sin^{-1}(x)$</i> $= \frac{\pi}{2} + 2\left(-\frac{\pi}{2}\right)$ <i>or consider range $[-\frac{\pi}{2}, \frac{\pi}{2}]$</i> $= -\frac{\pi}{2} \therefore k = -\frac{1}{2}$	1 mark
	(ii) Domain of $\cos^{-1}(x - 1)$ is $-1 \leq x \leq 1$ $\therefore -1 \leq x - 1 \leq 1, 0 \leq x \leq 2$ Domain of $\sin^{-1}(x - 2)$ is $-1 \leq x \leq 1$ $\therefore -1 \leq x - 2 \leq 1, 1 \leq x \leq 3$ $\therefore x \in [1, 2]$ <i>overlap of domains.</i>	2 marks

(d)	Let $\alpha = \arctan \frac{4}{3}$ and $\beta = \arctan \frac{1}{2}$ $\frac{4}{3} = \tan \alpha, \sin \alpha = \frac{4}{5}, \cos \alpha = \frac{3}{5}$ $\frac{1}{2} = \tan \beta, \sin \beta = \frac{1}{\sqrt{5}}, \cos \beta = \frac{2}{\sqrt{5}}$ $\therefore \sin \left(\arctan \frac{4}{3} + 2 \arctan \frac{1}{2} \right)$ $= \sin(\alpha + 2\beta)$ <i>note correct algebra.</i> $= \sin \alpha \cos 2\beta + \cos \alpha \sin 2\beta$ <i>use reference sheet for correct expansion</i> $= \left(\frac{4}{5} \right) (2 \cos^2 \beta - 1) + \left(\frac{3}{5} \right) (2 \sin \beta \cos \beta)$ $= \left(\frac{4}{5} \right) \left(2 \times \left(\frac{2}{\sqrt{5}} \right)^2 - 1 \right) + \left(\frac{3}{5} \right) \left(2 \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}} \right)$ $= \left(\frac{4}{5} \times \frac{3}{5} \right) + \left(\frac{3}{5} \times \frac{4}{5} \right)$ $= \frac{24}{25}$	3 marks
(e)		2 marks

$$\tan \beta = \frac{1}{2}, \tan \alpha = \frac{1}{3}$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}}$$

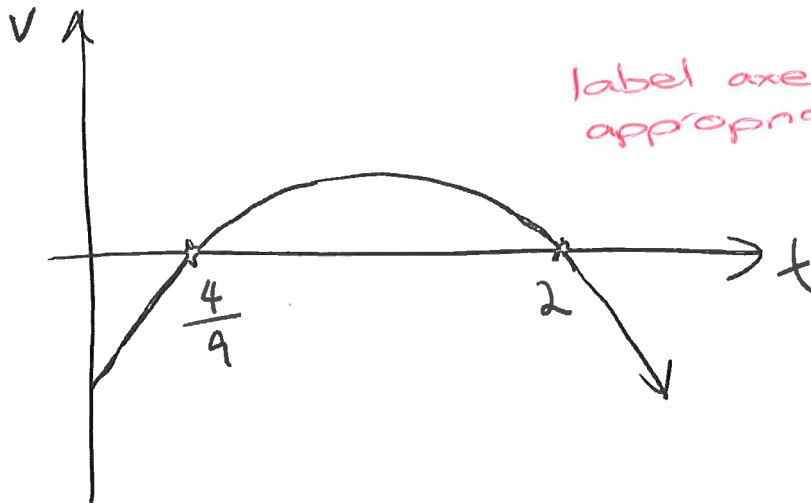
$$= \frac{5}{6} \div \frac{5}{6}$$

$$= 1$$

$$\therefore \alpha + \beta = \tan^{-1}(1) = \frac{\pi}{4}$$

show question
so write full solution.

(f)



label axes
appropriately.

2 marks